

TAEHYEON KIM

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SUMMARY

AI researcher in agent learning, reinforcement learning, and efficient reasoning. Senior Member of Technical Staff at Holiday Robotics, developing dense-feedback harness learning, text-and-motion-conditioned RL policies, and embodied-agent benchmarks. Previously led agentic search and LLM alignment at LG AI Research. My research spans adaptive thinking, controlled decoding, and inference acceleration.

SELECTED RESEARCH CONTRIBUTIONS

Root-token Policy Optimization (RPO) [C15] – first author, **ACL 2026 (Oral)**. [Paper](#)

- Proposed an RL framework that uses group-relative rewards to optimize the initial root-token decision: whether to produce an explicit reasoning trace.
- Learned difficulty-aware adaptive thinking at approximately **2% of the training compute** of the prior adaptive-reasoning methods evaluated, across multiple model families and scales.

Instructive Decoding [C7] – co-first author, **ICLR 2024 Spotlight**. [Paper](#)

- Proposed a training-free method that contrasts next-token predictions with predictions from noisy instructions to improve instruction following, without updating model parameters.

Blockwise parallel decoding with draft refinement [C10] – first author, **NeurIPS 2024**. [Paper](#)

- Analyzed predictions across multiple token heads and proposed draft-refinement algorithms using n-gram and neural language-model rescore.
- Increased mean accepted token length by **5–25% relative**, yielding **over 3x wall-clock speedup** over standard autoregressive decoding on the evaluated open-source 7B and 13B LLMs.

Additional selected work

- **Contrastive Thinking Decoding [C16]** – first author, **NeurIPS 2026**. Steering answer generation in reasoning models.
- **Multi-Drafter Speculative Decoding [C14]** – co-first author, **Findings of ACL 2026**. Multiple drafters with alignment feedback for speculative decoding.
- **Knowledge distillation [C2]** – co-first author, **IJCAI 2021**. Comparing Kullback-Leibler divergence and mean squared error objectives for knowledge distillation.

WORK EXPERIENCE

Holiday Robotics

Jun 2026 - Present

Senior Member of Technical Staff, Embodied AI Team

- **Harness learning:** Developing meta-harness-style training with dense feedback to guide the learning of agent execution harnesses. Founding member of the Embodied AI Team.
- **Text-and-motion-conditioned RL policies:** Developed an algorithm that automates manual engineering in constructing RL action policies from text and motion. Behavior-cloning regularization helps the resulting policies follow intended behaviors.
- **Evaluation (in progress):** Building benchmarks for embodied-agent behavior and task execution.

LG AI Research

Apr 2025 - Jun 2026

Superintelligence Lab: Research Scientist (Apr-Aug 2025); Squad Lead (Sep-Dec 2025); Tech Lead (Jan-Jun 2026)

- **Research leadership:** Led a team of **10+ researchers (6+ PhDs)** on agentic search and LLM alignment; defined the research roadmap, mentored researchers, and coordinated project delivery.
- **Agentic search:** Built an end-to-end search system over internal expert documents, from retrieval to agentic reasoning. Achieved higher accuracy than Perplexity, Codex, and Anthropic systems in internal evaluations on this task.
- **Agent systems and multimodal learning:** Built multi-tool orchestration with parallel retrieval, query planning, and examination tools. Trained dense retrievers and rerankers with hard-negative mining; designed VLM experiments for diagram understanding and retrieval grounding.
- **LLM alignment and RL:** SFT, DPO, and GRPO-based training; first-author RPO (ACL 2026, Oral [C15]); small-large model collaboration; Masked-SIMPER (in preparation [U3]).
- **Research:** Contrastive Thinking Decoding (NeurIPS 2026 [C16]); co-corresponding author on AdaSTaR (NeurIPS 2025 [C12]); MERIT negotiation benchmark (ACL 2026 [C13]); multi-drafter speculative decoding (Findings of ACL 2026 [C14]).

Google Research

Oct 2023 - Dec 2023

PhD Research Intern; hosts: Adrian Benton, Ananda Theertha Suresh, and Kishore Papineni

- Developed fast parallel decoding methods for LLMs; first-author NeurIPS 2024 paper [C10].

Dynamo AI (YC-backed)

Jan 2023 - May 2023

PhD Research Intern; worked with Eric Lin

- Privacy-preserving federated learning for enterprise AI; contributed to automotive B2B contracts and a first-author NeurIPS 2023 paper [C5].

National Institute of Meteorological Sciences (NIMS)

Mar 2022 - Jun 2023

Research Manager

- Transformer-based precipitation forecasting; NeurIPS 2022 Weather4Cast competition award, collaborating with a team of 40+ researchers.

Qualcomm AI

Jun 2021 - Dec 2021

PhD Research Intern; worked with Heesoo Myeong

- Resource-efficient backbone design for ADAS; ICML 2022 HAET Workshop [W2] and U.S. patent [P1].

EDUCATION

Ph.D., Graduate School of AI, KAIST

Mar 2020 - Feb 2025

Advisor: Prof. Se-Young Yun

Thesis: Efficient and Effective Inference-time Decoding Strategies for Foundational Models.

M.S., Graduate School of Data Science, KAIST

Mar 2018 - Feb 2020

Advisor: Prof. Se-Young Yun

Thesis: Orthogonal Feature Regularization: A Novel Approach for Training Robust Models.

B.S., Mathematical Sciences, KAIST

Mar 2013 - Feb 2018

Minor: Intellectual Property Program

PUBLICATIONS

* equal contribution; † corresponding authors.

- [C16] Kim, T., Jang, Y., Lee, H., Kim, Y. J. & Lee, M. (2026). Contrastive Thinking Decoding: Steering Answer Generation in Reasoning Models, **NeurIPS 2026**.
- [C15] Kim, T., Lee, H., Jang, Y. & Lee, M. (2026). Efficiently Learning To Reason or Not to Reason: Root-token Policy Optimization for Adaptive Thinking, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**) (Oral), Jul. 2026.
- [C14] Kim, T., Jung, H.* & Yun, S. (2026). Multi-Drafter Speculative Decoding with Alignment Feedback, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026 Findings**), Jul. 2026.
- [C13] Oh, J., Aghazada, M., Shin, Y., Yun, S. & Kim, T.† (2026). MERIT Feedback Elicits Better Bargaining in LLM Negotiators, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**), Jul. 2026.
- [W9] Kim, Y., Yi, E., Kim, M., Yun, S.†, Kim, T.† (2025). Guiding Reasoning in Small Language Models with LLM Assistance, **CoLM 2025 Workshop: Test-time Scaling and Reasoning Models**.
- [C12] Koh, W., Oh, W., Jang, J., Lee, M., Kim, H., Kim, A., Kim, J., Lee, J., Kim, T.†, Yun, S.† (2025). AdaSTaR: Adaptive Data Sampling for Training Self-Taught Reasoners, *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, Dec. 2025.
- [C11] Koh, W., Yoon, J., Lee, M., Song, Y., Cho, J., Kang, J., Kim, T., Yun, S., Yu, Y., Lee, B. (2025). C²: Scalable Auto-Feedback for LLM-based Chart Generation, *The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL 2025)* (Oral), May. 2025.
- [J2] Kim, T., Kim, D. & Yun, S. (2024). FLR: Label-Mixture Regularization for Federated Learning with Noisy Labels, *Transactions on Machine Learning Research (TMLR)*.
- [W8] Kim, T., Jung, H.* & Yun, S. (2024). A Unified Framework for Speculative Decoding with Multiple Drafters as a Bandit, **NeurIPS 2024 Workshop: Efficient Natural Language and Speech Processing (ENLSP-IV)**.
- [C10] Kim, T., Suresh, A.T., Papineni, K., Riley, M., Kumar, S. & Benton, A. (2024). Accelerating Blockwise Parallel Language Models with Draft Refinement, *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, Dec. 2024.
- [C9] Ho, N.*, Bae, S.*, Kim, T., Jo, H., Kim, Y., Schuster, T., Fisch, A., Thorne, J. and Yun, S. (2024). Block Transformer: Global-to-Local Language Modeling for Fast Inference, *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, Dec. 2024.
- [C8] Yi, E.*, Kim, T.*, Jeung, H., Chang, D. and Yun, S. (2024). Towards Fast Multilingual LLM Inference: Speculative Decoding and Specialized Drafters, *The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*, Nov. 2024.

- [W7] Kim, T., Suresh, AT., Papineni, K., Riley, M., Kumar, S. & Benton, A. (2024). Exploring and Improving Drafts in Blockwise Parallel Decoding, ICML 2024 Workshop: Efficient Systems for Foundation Models (Es-FoMo).
- [C7] Kim, T.*, Kim, J.*, Lee, G.* & Yun, S. (2024). Instructive Decoding: Instruction-Tuned Large Language Models are Self-Refiner from Noisy Instructions, *Twelfth International Conference on Learning Representations (ICLR 2024) (Spotlight, Top 5%)*, May. 2024.
- [C6] Yang, Y., Kim, T. & Yun, S. (2024). Leveraging Normalization Layer in Adapters with Progressive Learning and Adaptive Distillation for Cross-Domain Few-Shot Learning, *The 38th Annual AAAI Conference on Artificial Intelligence (AAAI 2024)*, Feb. 2024.
- [C5] Kim, T., Lin, E., Lee, J., Lau, C. & Mugunthan, V. (2023). Navigating Data Heterogeneity in Federated Learning: A Semi-Supervised Federated Object Detection, *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, Dec. 2023.
- [C4] Gruca et al. Weather4cast at NeurIPS 2022: Super-Resolution Rain Movie Prediction under Spatio-temporal Shifts, *Proceedings of the NeurIPS 2022 Competition and Demonstration Track*, in *Proceedings of Machine Learning Research 220:292-313 (2022)*.
- [W6] Kim, T., Kang, S., Shin, H., Yoon, D., Eom, S., Shin, K. & Yun, S. (2022). Region-Conditioned Orthogonal 3D U-Net for Weather4Cast Competition, NeurIPS 2022 Workshop: Weather4Cast Competition.
- [W5] Eom, S., Kim, T. & Yun, S. (2022). Layover Intermediate Layer for Multi-Label Classification in Efficient Transfer Learning, NeurIPS 2022 Workshop: Has it Trained Yet? (HITY).
- [W4] Shin, J., Kim, T. & Yun, S. (2022). Revisiting the Activation Function for Federated Image Classification, NeurIPS 2022 Workshop Federated Learning.
- [W3] Kim, T. & Yun, S. (2022). Supernet Training for Federated Image Classification under System Heterogeneity, ICML 2022 Workshop on Dynamic Neural Networks (Oral).
- [J1] Kim, T. & Yun, S. (2022). Revisiting Orthogonality Regularization: A Study for Convolutional Neural Networks in Image Classification, *IEEE Access*, Jun. 2022.
- [W2] Kim, T., Myeong, H. & Yun, S. (2022). Revisiting Architecture-aware Knowledge Distillation: Smaller Models and Faster Search, ICML 2022 Hardware Aware Efficient Training (HAET) Workshop, July 2022.
- [C3] Kim, T.*, Ko, J.*, Cho, S., Choi, J. & Yun, S. (2021). FINE Samples for Learning with Noisy Labels, *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, Dec. 2021.
- [C2] Kim, T.*, Oh, J.*, Kim, N., Cho, S., & Yun, S. Y. (2021). Comparing Kullback-Leibler Divergence and Mean Squared Error Loss in Knowledge Distillation. In *the 30th International Joint Conference on Artificial Intelligence (IJCAI 2021)*, Aug. 2021 (acceptance rate: 13.9%)
- [W1] Kim, T.*, Ahn, J.*, Kim, N.*, & Yun, S. (2020). Adaptive Local Bayesian Optimization Over Multiple Discrete Variables. Workshop at NeurIPS 2020 Competition Track on Black-Box Optimization Challenge, Dec. 2020.
- [C1] Kim, T., Kim, J. & Yun, S. (2019). Efficient Model for Image Classification With Regularization Tricks. *Proceedings of the NeurIPS 2019 Competition and Demonstration Track*, in *Proceedings of Machine Learning Research 123:13-26*.
- [T] Kim, T.. Orthogonal feature regularization : a novel approach for training robust models, Korea Advanced Institute of Science and Technology (KAIST)

ONGOING MANUSCRIPTS

- [U3] Kim, T. et al. (2025). Masked-SIMPER: Improving Collaborative Reasoning, In preparation.
- [U1] Bartholet, M.*, Kim, T.*, Beuret, A. Yun, S. & Buhmann, J. (2024). Hypernetwork-Driven Model Fusion for Federated Domain Generalization, Under review.

US PATENTS

- [P1] Kim, T., Myeong, H. (2023). Trust-Region Aware Architecture Distillation for Sample-Efficient Neural Architecture Search, Qualcomm Inc.

AWARDS & HONORS

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|---|----------|
| • Gold Reviewer (top 25%), ICML 2026 | Jul 2026 |
| • Google Conference Scholarship, NeurIPS 2024 draft-refinement paper | Dec 2024 |
| • Google Cloud Platform Credit Award, Gemma Academic Program | Mar 2024 |
| • 4th Place, NeurIPS Weather4Cast Competition; Project Lead | Dec 2022 |
| • Qualcomm AI Fellowship | Oct 2022 |
| • Best Poster Awards, KAIST AI Workshop (FINE; knowledge distillation) | Jan 2022 |
| • 8th Place, NeurIPS Black-Box Optimization Challenge; Project Lead | Dec 2020 |
| • 2nd & 3rd Places, NeurIPS MicroNet Challenge, CIFAR-100; Project Lead | Dec 2019 |

LEADERSHIP

Doctoral Student Representative, KAIST AI

Mar 2021 - Feb 2022

Built student organizations and served as liaison between faculty and students.

Lab Master, OSI Lab, KAIST

Mar 2020 - Feb 2021

Organized seminars, lab policy, and funding processes; liaison between advisor and students.

Vice Captain, ELKA (KAIST cheerleading group)

Oct 2014 - Nov 2015

KAIST festival planning and promotion; 1st prize at the 2015 Gwangju Summer Universiade cheering festival.

INVITED TALKS

- Naver AI Lab Seminar – LLM Inference-Time Strategies and Beyond Jan 2025
- Microsoft Research Seminar – Instructive Decoding and Beyond Jan 2025
- Squeezebits Efficient AI Seminar – Towards Fast LLM Inference Jun 2024
- KCC 2024 – Semi-Supervised Federated Object Detection Jun 2024
- HyperConnect Seminar – Personalized Federated Learning under Label Noise Jan 2023
- KSC 2022 – FINE Samples for Learning with Noisy Labels Dec 2022
- Naver Clova AI Seminar – Orthogonal Feature Regularization Jun 2019

SERVICE & SKILLS

- **Reviewer:** NeurIPS 2022-2024; ICML 2022-2024, 2026; ICLR 2024-2025; AutoML-Conf 2022; ECCV 2022; KDD 2023; ARR 2024. Journal: Pattern Recognition.
- **Skills:** Python, PyTorch, LaTeX, vLLM/SGLang. LLM post-training (SFT, DPO, GRPO), inference optimization, retrieval, and agent tool orchestration.
- **AI in art:** Surplus Human, with Vania Oh and Junbeom Shin. AI with Weird Wonderland, CELINE PARK GALLERY, March 2022; organized by Next Interface Lab.

TEACHING & MENTORING

- AI Lecturer, ELICE – Mathematics in ML; lectures for LG employees Jan-Feb 2022
- AI Mentor, Curinc – Introduction to Deep Learning; 70-hour course Jun-Jul 2021
- TA, LG – Computer Vision and Deep Learning Oct-Nov 2020
- Advisory Committee, National Science Museum – Deep Learning Mar-Oct 2020
- TA, KAIST Data Science – Introduction to Deep Learning Mar-Jul 2019
- TA, LG – Computer Vision and Deep Learning Jan 2019
- Lecturer, Samsung – Python Basics Jan 2019
- Lecturer, KAIST – Regular Leadership Course Mar-Dec 2017

SCHOLARSHIPS & FELLOWSHIPS

- KAIST Graduate School of AI doctoral funding: US\$20,000 per year during Ph.D. studies.
- Korea Student Aid Foundation graduate scholarship: US\$16,000 per year Mar 2018 - Feb 2020
- KAIST full academic scholarship: US\$3,000 per year Mar 2013 - Aug 2017